



GIS & Remote Sensing Concepts

(Based on Project 1 – Drought Severity Mapping & Project 2 – Urban Heat Island Analysis)

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1. Remote Sensing

What is Remote Sensing?

Remote Sensing is the science of collecting information about the Earth's surface **without physically visiting the location**. It uses satellites or aircraft to observe and analyze different features of the Earth.

Simple Explanation

Instead of travelling to Bundelkhand or Delhi to collect information manually, satellites observe these places from space and provide data that can be analyzed later.

Why is it Important?

- Saves time.
 - Covers very large areas.
 - Provides regular observations.
 - Useful for agriculture, forests, disasters, climate studies and urban planning.
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2. How Does a Satellite Observe the Earth?

A satellite does **not** directly "see" the Earth.

It records the **light reflected** or **heat emitted** from the Earth's surface.

Workflow

Sunlight

↓

Falls on Earth

↓

Different objects reflect different amounts of light

↓

Satellite records this reflected light

↓

Computer converts it into an image

↓

Analysis is performed

Remember:

Satellite records reflected (or emitted) radiation, not the object itself.

3. Electromagnetic Radiation (EMR)

Electromagnetic Radiation is simply **energy that travels in the form of waves**.

The Sun continuously sends this energy towards the Earth.

Satellites use this energy to observe different objects.

Humans can only see **Visible Light**, whereas satellites can also detect:

- Near Infrared (NIR)
- Shortwave Infrared (SWIR)
- Thermal Infrared
- Other wavelengths

This is why satellites can detect information that our eyes cannot.

Remember:

Satellites can "see" much more than human eyes.

4. Reflection and Absorption of Light

When sunlight falls on any object,

some energy is

- absorbed,
- reflected,
- transmitted.

Different objects reflect different amounts of light.

Example:

Healthy vegetation

✓ Absorbs Red light

✓ Reflects Near Infrared (NIR)

Water

✓ Absorbs most wavelengths

Buildings

✓ Reflect differently than vegetation

This difference helps satellites identify different objects.

Remember:

Every object reflects light differently.

5. Spectral Signature

Every object has its own unique pattern of reflecting and absorbing light.

This unique reflection pattern is called its **Spectral Signature**.

Just like every person has different fingerprints,
every object has a different spectral signature.

Example:

Water → Very low NIR reflection

Healthy vegetation → Very high NIR reflection

Built-up areas → High SWIR reflection

Why is it Important?

Without spectral signatures,
the satellite would not be able to distinguish forests, water, soil or buildings.

Remember:

Spectral Signature = Fingerprint of an object.

6. Satellite Bands

A satellite does not capture only one image.

Instead,

it records different wavelength ranges separately.

Each wavelength range is called a **Band**.

Every band provides different information about the Earth's surface.

Bands Used in Our Projects

Band 4 (Red)

Used for vegetation analysis.

Band 5 (Near Infrared - NIR)

Healthy vegetation strongly reflects this band.

Band 6 (SWIR)

Useful for identifying built-up areas.

Band 10 (Thermal Infrared)

Used to estimate Land Surface Temperature.

Remember Forever

A Band is a specific wavelength range recorded by a satellite sensor.

7. Pixel

A satellite image is made up of thousands or millions of tiny squares.

Each square is called a **Pixel**.

For Landsat imagery,

one pixel represents approximately

30 m × 30 m on the Earth's surface.

Every pixel stores its own information.

Example:

Pixel A

NDVI = 0.72

Pixel B

NDVI = 0.28

This means every location has its own vegetation condition.

Remember:

Each pixel has its own value.

8. Raster Data

Raster data is made up of pixels arranged in rows and columns.

Each pixel contains a numerical value representing a property such as:

- Vegetation
- Temperature
- Elevation
- Moisture

Satellite images are raster datasets.

Used In

Project 1

Project 2

Remember:

Raster = Grid of Pixels.

9. Area of Interest (AOI)

The Area of Interest (AOI) is the specific region selected for analysis.

Instead of processing the whole Earth,
we only analyze the required study area.

Example

Project 1

AOI = Bundelkhand

Project 2

AOI = Delhi NCR

Why is AOI Important?

- Faster processing.
 - Lower computation.
 - More focused analysis.
-

Remember Forever

AOI tells the software exactly where analysis should be performed.

10. Time-Series Analysis

Time-Series Analysis means studying the **same location over different time periods**.

Instead of analyzing only one image,
multiple years of satellite data are compared.

This helps us understand

- historical condition,
- current condition,
- seasonal changes,
- long-term trends.

In Project 1,
we analyzed NDVI from **2015–2025**.

Why is it Important?

One image only tells the present.

Multiple years tell the complete story.

Remember:

Time-Series = Same place + Different times.

11. NDVI (Normalized Difference Vegetation Index)

NDVI measures the **health of vegetation**.

Healthy plants

✓ Reflect high Near Infrared (NIR)

✓ Absorb Red light

Dry plants

✓ Reflect less NIR

✓ Reflect relatively more Red

Formula

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Why this Formula?

NIR – Red

Shows vegetation contrast.

NIR + Red

Normalizes values so they remain comparable.

Interpretation

$NDVI < 0$

Water

$0 - 0.2$

Bare Soil

$0.2 - 0.5$

Sparse Vegetation

0.5

Healthy Vegetation

Remember:

NDVI tells the current health of vegetation, not drought directly.

12. VCI (Vegetation Condition Index)

VCI compares the **current vegetation condition** with its **historical minimum and maximum conditions**.

Unlike NDVI,

VCI tells whether vegetation is performing normally or under stress.

Why Do We Need VCI?

A low NDVI in Rajasthan may be normal.

The same NDVI in Punjab may indicate drought.

VCI removes this problem by comparing the current NDVI with the historical record of the same pixel.

Interpretation

$VCI < 0.2$

Extreme Drought

$0.2 - 0.4$

Moderate Drought

$0.4 - 0.6$

Mild Drought

0.6

Healthy Vegetation

Remember:

NDVI shows health. VCI compares today's health with history.

13. Google Earth Engine (GEE)

Google Earth Engine is Google's cloud platform for satellite data analysis.

Instead of downloading large satellite datasets,

we write code,

and Google performs the calculations on its own servers.

Advantages

- No need to download huge datasets.
- Very fast processing.
- Access to global satellite archives.
- Suitable for large-scale analysis.

Remember:

Google Earth = Visualization

Google Earth Engine = Analysis

14. Concept Flow of Project 1

Study Area (Bundelkhand)

↓

Satellite Data (MODIS NDVI)

↓

Time-Series Analysis

↓

NDVI

↓

Historical NDVI Minimum

↓

Historical NDVI Maximum

↓

VCI Calculation

↓

Drought Classification

↓

Area Statistics

↓

Final Drought Severity Map

15. Concept Flow of Project 2

Study Area (Delhi NCR)



Landsat-8 Satellite Data



Band Selection



NDVI



NDBI



Land Surface Temperature (LST)



Urban Heat Island Identification



Area Statistics



Final UHI Map

✚ Final Revision (30-Second Memory Map)

Remote Sensing



Satellite



EMR (Sunlight)



Reflection



Spectral Signature



Bands



Pixels



Raster Image



AOI Selection



Google Earth Engine



Time-Series



NDVI



VCI (Project 1)



LST + NDBI (Project 2)



Final Maps & Area Statistics